

# Kalhara Tennakoon

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## PROFILE

First Class Computer Science graduate and Software Engineering batch top with hands-on industry experience in cloud-native development, Kubernetes, CI/CD, and cloud security at IFS R&D International. Experienced in building full-stack applications, ML forecasting systems, and AI applications using local LLMs, RAG, and vector search. Seeking Software Engineering, DevOps, Cloud, or MLOps opportunities where I can build reliable and intelligent systems.

## EDUCATION

### Kingston University, London, UK

BSc (Hons) Computer Science (Software Engineering) — First Class Honours

2026

- Batch Top – Software Engineering, ESU Kandy

## EXPERIENCE

### IFS R&D International (Pvt) Ltd

Colombo, Sri Lanka

DevOps Engineering Intern

Jul 2024 – Oct 2024

- Automated Kubernetes-native CI/CD workflows using Tekton, streamlining deployment processes for critical services.
- Integrated Behave automated testing into Tekton pipelines, reducing manual debugging effort by approximately 30%.
- Optimized pipeline configurations and resolved recurring build and deployment issues to improve deployment reliability.
- Conducted knowledge-sharing sessions on testing best practices to support team collaboration and code quality.

Software Engineering Intern

Jul 2020 – Dec 2020

- Developed a Docker/Kubernetes-based security scanning solution for container platforms and vulnerability assessment workflows.
- Integrated security scanning tools into CI/CD pipelines to improve vulnerability detection across deployment stages.
- Diagnosed deployment and runtime issues and authored technical documentation supporting system reliability and onboarding.

## PROJECTS

### VetCare Pro — Smart Veterinary Clinic Management System | Grade A (76%)

React, Node.js, Express, PostgreSQL, pgvector, Flask, Scikit-learn, Ollama, SwiftUI

Oct 2025 – Aug 2026

- Built a full-stack veterinary clinic platform covering appointments, medical records, billing, inventory, role-based access, and JWT authentication.
- Extended the platform with a local RAG assistant using Ollama and pgvector, combining semantic retrieval with exact SQL queries and human confirmation for database writes.
- Added ML forecasting models, a pet-owner portal, and SwiftUI companion iOS app; selected among the Top 10 of Ascentic AI Launch Pad 2026.

### SolarCast — Personal Solar Energy Forecasting App

Python, FastAPI, React, Prophet, Pandas, REST APIs

Apr 2026 – Present

- Built a full-stack ML application predicting solar energy generation, grid export, and cash payouts from residential inverter data.
- Developed a time-series forecasting pipeline using Meta Prophet with 90% confidence interval predictions and chronological train/test splitting to prevent data leakage.
- Built an interactive React dashboard with CSV upload and energy analytics visualizations.

### Predicting Course Difficulty from Student Evaluation Responses

Python, Scikit-learn, Pandas, Matplotlib, Seaborn, SPSS

Feb 2025 – Jul 2025

- Developed supervised ML models to predict perceived course difficulty from student evaluation datasets.
- Performed data preprocessing, exploratory analysis, feature engineering, and model evaluation using cross-validation.
- Applied statistical significance testing and model interpretation to identify instructor-related attributes associated with perceived course difficulty.

### Kubernetes Cluster & Container Image Security Scanner

Kubernetes, Docker, Golang, Argo CD, Rancher, Azure DevOps

Jul 2020 – Dec 2020

- Developed a container security scanning solution for Kubernetes environments, detecting vulnerabilities in container images and cluster configurations.
- Integrated automated security scanning workflows into CI/CD pipelines to strengthen secure deployment practices.

HONOURS & AWARDS

|                                                                                                                        |             |
|------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Top 10 Finalist – Ascentic AI Launch Pad</b>                                                                        | 2026        |
| Selected to present VetCare Pro at Demo Day for its local RAG assistant, ML forecasting models, and companion iOS app. |             |
| <b>Batch Top – Software Engineering, ESU Kandy</b>                                                                     | 2026        |
| Ranked first in the Software Engineering cohort.                                                                       |             |
| <b>Two-Time Dan Kohn Scholarship Recipient – KubeCon + CloudNativeCon Europe</b>                                       | 2022 & 2025 |
| Awarded scholarships recognising contributions to the cloud-native ecosystem.                                          |             |
| <b>First Runner-Up – Hack:Bit Hackathon</b>                                                                            | 2019        |
| Secured 2nd place in a competition organised in collaboration with Microsoft and Sarvodaya Fusion.                     |             |

TECHNICAL SKILLS

**Programming:** Python, Java, JavaScript, Go, C, Bash/Shell  
**AI / ML:** Scikit-learn, Pandas, Prophet, pgvector, Ollama, RAG, Data Cleaning, EDA  
**Cloud & DevOps:** Kubernetes, Docker, Tekton, Argo CD, GitHub Actions, Azure DevOps, CI/CD, Infrastructure as Code  
**Cloud Platforms:** Microsoft Azure, AWS, Google Cloud Platform, AKS, GKE, Rancher  
**Backend & Web:** Node.js, Express, React, Flask, FastAPI, REST APIs  
**Databases:** PostgreSQL, MySQL, MongoDB  
**Tools & Practices:** Git, GitHub, Jira, Agile, Scrum, Technical Documentation

CERTIFICATIONS

|                                                         |      |
|---------------------------------------------------------|------|
| <b>Cloud / DevOps</b>                                   |      |
| • Containers & Kubernetes Essentials — IBM              | 2025 |
| • GitHub Foundations — GitHub                           | 2025 |
| • Introduction to Kubernetes — Linux Foundation         | 2024 |
| • Microsoft Certified: Azure Fundamentals (AZ-900)      | 2023 |
| <b>AI / Data</b>                                        |      |
| • Google AI Professional Certificate                    | 2026 |
| • Microsoft Certified: Azure Data Fundamentals (DP-900) | 2024 |
| • Microsoft Certified: Azure AI Fundamentals (AI-900)   | 2024 |

TECHNICAL WRITING

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|-------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Technical Writer — Medium</b>                                                                                        | Dec 2021 – Present |
| • Authored 50+ technical articles on DevOps, cloud-native technologies, and software engineering, reaching 150K+ views. |                    |
| • Published technical content covering Kubernetes, Tekton, Docker, Go, and CI/CD through Level Up Coding and FAUN.dev.  |                    |